

Sentinel Fusion – Team Work Allocation (4 Members)

Objective: Split the project into four independent workstreams so each team member owns a major subsystem while collaborating during integration.

Ashwin Manoj — Frontend & UI

Responsibilities

- Next.js setup
- Authentication screens
- Dashboard
- Evidence upload UI
- Case management
- Timeline view
- Relationship graph
- Risk dashboard
- Report viewer
- API integration

Tech Stack

- Next.js
- React
- Tailwind CSS
- shadcn/ui
- React Flow
- Cytoscape.js
- Apache ECharts

Deliverables

- Complete frontend
- Responsive UI
- Integrated APIs

Abhinand K — Backend & Database

Responsibilities

- FastAPI backend
- REST APIs
- PostgreSQL schema

- Neo4j graph model
- Redis integration
- File upload
- Authentication
- OpenAPI docs

Tech Stack

- FastAPI
- PostgreSQL
- Neo4j
- Redis
- SQLAlchemy

Deliverables

- Backend services
- Database
- API documentation

Joyal Blesson — AI & Intelligence

Responsibilities

- OCR
- Metadata extraction
- Whisper transcription
- Image understanding
- Deepfake detection
- Risk scoring
- Timeline reconstruction
- Relationship extraction
- LangGraph workflows
- Report generation

Tech Stack

- Python
- LangGraph
- GPT-5.5 API
- PaddleOCR
- Whisper
- OpenCV
- ExifTool

- Qdrant

Deliverables

- AI pipelines
- AI APIs
- Model evaluation

Rosemary Rejimon — DevOps, Security & Integration

Responsibilities

- Docker
- CI/CD
- Deployment
- Monitoring
- Logging
- Object storage
- RBAC
- Audit logs
- Performance testing
- Final integration
- Documentation

Tech Stack

- Docker
- GitHub Actions
- MinIO/S3
- Prometheus
- Grafana
- Loki
- Sentry

Deliverables

- Production deployment
- Infrastructure
- Security configuration

Shared Responsibilities

- Sprint planning
- Code reviews

- Git pull requests
- Bug fixing
- Testing
- Documentation
- Final presentation

Repository Structure

```
frontend/  
backend/  
ai/  
infra/  
docs/  
docker/
```

Sprint Plan

Sprint 1: Frontend setup, Backend setup, AI prototype, Infrastructure setup

Sprint 2: Core feature development

Sprint 3: Integration, Testing, Bug fixing

Sprint 4: Final polish, Documentation, Deployment, Demo

Branch Strategy

- main
- develop
- feature/frontend
- feature/backend
- feature/ai
- feature/devops

Workflow: Feature → Develop → Main